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Fifteenth Annual In-HcDelamater

Department of Psychology

Dalhousie University

24-25 April 1989

Room 4258/63

Fifteenth Annual In-house Convention

Department of Psychology

Dalhousie University

24-25 April 1989

Schedule

Monday, 24 April

11:00 - 12:00

Session I. Perception. Chair: E. Coscia

- ✓ 1. D.P. Phillips, S.E. Hall & J.L. Hollett
- ✓ 2. P. Kind & D.E. Mitchell
- ✓ 3. C.J. Beaver & S.E. Hall
- ✓ 4. D.E. Mitchell & S.E. Hall

LUNCH

1:30 - 3:15

Session II. Development. Chair: S. Wright

- 5. M.G. Yoon & M.J. Yoon
- 6. C. Moore & K. Pure
- 7. N. Akhtar, F.S. Dunham & P.J. Dunham
- 8. M.E. Farmer & S.E. Bryson
- 9. F. Rosenthal
- 10. J. Barresi

COFFEE BREAK

3:30 - 5:00

Session III. Animal Learning. Chair: D. Phillips

- 11. W.K. Honig
- 12. B.L. Alsop & W.K. Honig
- 13. D. Williams & V. LoLordo
- 14. A. Droungas, A.R. Delamater, & V. LoLordo
- 15. A.R. Delamater & V.M. LoLordo

BEER IN LOUNGE

Tuesday, 25 April

9:30 - 11:15

Session IV. Neuroscience. Chair: R. Klein

16. R.P. Croll & K.G. Bina
17. B. Rusak & H.A. Robertson
18. J.H. Brandstatter, I.A. Meinertzhagen, S.R. Shaw
19. I.A. Meinertzhagen & H. Seyan
20. S.R. Shaw
21. P. Kind, D.E. Mitchell, & C.J. Beaver

COFFEE BREAK

put up poster

11:30 - 12:30

Session V. Animal Behavior. Chair: G. Eskes

22. R.E. Brown, P.B. Singh, & B. Roser
23. H.A. Schellinck, V.C. Rouse, & R.E. Brown
24. C.A. Staicer
25. E.M. Coscia, D.P. Phillips, J.C. Fentress

LUNCH

12:15 - 2:00

Session VI. Posters.

26. C.L. Ryan & R.E. Brown
27. S.L. Wright & R.E. Brown
28. R.E. Brown & L.A. Wisker
29. M. Legerstee
30. J. Ryon, R.E. Brown, & J. Fentress
31. A. D'Aloisio & R. Klein

2:00 - 3:30

Session VII. Human Experimental. Chair: A. Droungas

32. A. Cohen
33. R. Klein, G. Berry, K. Briand, & B. D'Entremont
34. A. Kingstone
35. C.J. Smith
36. N. Phillips, J.F. Connolly, G. Eskes, C. Mate-Kole

COFFEE BREAK

4:00 - 5:00

Session VIII. Panel Discussion. Chair: C. Moore

"The Rushton Affair"

Speakers: R. Brown
J. Connolly
R. Rodger
T. Jukes

DINNER AND BEER IN LOUNGE

ABSTRACTS

1. D.P. Phillips, S.E. Hall and J.L. Hollett

Summertime Diversions of a Man and Two Women. Or, Why It's Important That Spikes be Precisely Timed

A much-favoured neural code for the frequency content of a sound is the temporal one. This theory proposes that each acoustic event in the stimulus generates a spike discharge in an auditory neuron, so that the time between spikes specifies the spectral and temporal content of the sound. There is abundant evidence in the auditory nerve not only that this temporal representation exists, but also that it is competent for frequency elements up to about 3500 Hz. The further question, however, is the extent to which this temporal code significantly penetrates the central auditory nervous system. This question is not unreasonable, because the greater the number of serial synapses in the pathway transmitting the time code, the greater must be the imprecision in spike timing at the end of the neural pathway. In studies conducted last summer, we obtained evidence on this issue from single cat cortical neurons, by studying the effects of placing increasing temporal demands on a neuron's spike response rate. Our data show that cortical neurons which entrain spikes to brief tones presented at low repetition rates (eg., 1 Hz) fail to do so for tones at higher repetition rates, even though the precision of their spike timing is more than capable of supporting perfect entrainment. This suggests that factors other than spike time precision limit the temporal representation of acoustic events in the cortex. Second, we provided evidence that even under the most optimal of stimulus conditions, cortical cells probably do not have sufficient precision of spike timing to support a temporal code for components in excess of about 100 Hz. This suggests to us that the cortex does not possess a representation of the fine time structure of a sound's waveform.

2. P. Kind and D.E. Mitchell

Cell sizes in the Lateral Geniculate Nucleus of the Cat Following Reverse Occlusion Regimens which Optimize Visual Recovery from Early Monocular Deprivation.

Monocular deprivation in early life causes the anatomical and physiological development of the central visual pathway of kittens to be severely impaired and results in visual deficits resembling those observed in severe forms of amblyopia. Following monocular deprivation from birth to six weeks of age, the cells in the lateral geniculate nucleus (LGN) which receive input from the deprived eye are 25-30% smaller in area than those cells which receive input from the non-deprived eye. The present research was designed to investigate further the duration of the critical periods in the kitten LGN during which these anatomical changes occur. The kittens were monocularly deprived from birth to six weeks of age and then received a period of either full-time or part-time reverse occlusion. In addition, we examined the effects of a period of binocular vision imposed after the critical period on the morphology of the LGN. Finally several kittens were either reverse sutured or monocularly deprived at three months of age. The results indicate that a period of binocular visual exposure can alter LGN cell areas even when imposed after the critical period. In addition, the anatomy of the LGN is not directly correlated with the final visual capacities of the cat, but instead may reflect the visual acuities of the two eyes at the age of three months.

3. C.J. Beaver and D.E. Mitchell

Cortical areas mediating behavioural recovery following early periods of monocular deprivation in kittens.

Previous work in the laboratory has shown that following termination of six weeks of reverse occlusion of kittens that had been monocularly deprived until six weeks of age there is a surprising change in the visual acuities of both eyes. Although the visual acuity of the initially deprived eye

improves substantially during the period of reverse occlusion, during the subsequent binocular period the vision of the eye deteriorates rapidly while that of the initially non-deprived eye regains almost normal acuity. As of yet, the mechanism mediating these changes is still unknown. Lesions of areas 17 and 18 are being carried out to determine the location of the mechanism. Possible outcomes and their meaning will be discussed.

4. D.E. Mitchell and S. Hall

A Comparison of two Methods of Assessment of the Visual Acuity of Cats: A Search for Evidence of Aliasing.

The ability to resolve grating patterns at a given retinal location has traditionally been thought to be limited by the spacing of photo-receptors or sampling elements at that location. However, it is possible for acuity to exceed this limit in certain situations because of spurious resolution of patterns generated by undersampling of gratings by a regular array of sampling elements. The term "aliasing" is applied to such spurious resolution since the pattern observed has a spatial frequency different (lower) than that of the pattern that is actually imaged on the retina. Because of the potential for aliasing in the cat we have compared estimates of acuity obtained on a detection task (subject to aliasing) with estimates derived on a discrimination task that should be immune to aliasing effects. Since the two estimates were very similar it was concluded that conventional measures of acuity in the cat are not contaminated by aliasing.

5. M.G. Yoon and Myonghae Joy Yoon

Development of Narrative Discourse in Normal Children

We study the development of narration in the way how normal children represent an input-story into their own narrative outputs. Child-subjects (2.5 - 8 years old) and their mothers are invited to converse over an internal telephone system across two separate test rooms, while they can view each other via a reciprocal video-monitoring system. The mother reads a short story to the child over the phone as many times as the child wants. When the child is ready, he/she retells the story to a third party, the experimenter. All utterances of the child and the experimenter are recorded on videotapes, and later transcribed verbatim for detailed analyses. The utterances are segmented into unitary propositions with differential symbolic notations. Individual children's recounts of each story are depicted into characteristic textural graphs: unitary propositions constitute nuclear nodes which are interconnected with various cohesive ties into a semantically coherent pattern. These textural graphs are analyzed with respect to the categorical spectra of recalled propositions, modifications of syntactic structures, the pattern of cohesive ties, and the types of semantic inferences as compared with the template graph of the input-text. The results show that preschool children recalled action-propositions at higher frequencies whereas elementary school children recalled external state-propositions at higher rates. Few children recalled narrator's perspectives. The modes of syntactic combination of propositions became more elaborated as they grew older. The degree of causal interpretations inferred from the input-propositions also increased with age.

6. Chris Moore and Kiran Pure

Children's understanding of the expression of uncertainty

Modal terms (e.g. *must*, *might*, *probably*, *possibly*) are commonly used in natural language to express different degrees of the mental state of uncertainty. This paper reports an experiment which examined the development of children's understanding of this function of modals. Sixty children between the ages of three and six were presented with a task in which they had to find the location of an object hidden in one of two boxes. Their only clues were two statements said to them by two puppets. The two statements differed only in terms of the particular modal verb (of *must*, *might*,

and *could*) that was employed in the sentence frame "It [verb] be in the red/blue box" or modal adjunct (of *probably* and *possibly*) employed in the sentence frame "It's [adjunct] in the red/blue box." Results revealed that children's understanding of the distinctions between the modal terms with respect to certainty and uncertainty develops at around four to five years of age. This finding is in agreement with results from studies on mental terms such as *know* and *think*, and implies that children's understanding of the mental state concept of uncertainty develops at about this age.

7. Nameera Akhtar, Frances S. Dunham, and Philip J. Dunham

Directive interactions and early vocabulary development: The role of joint attentional focus.

Several dimensions of maternal speech correlate with aspects of language acquisition in the second year of life. Directiveness, for example, when assessed by syntactic or pragmatic aspects of the mother's speech (i.e. frequency of prescriptives), predicts slow vocabulary development. As prescriptives are most often used to direct a child's attention to a different activity or object, it may be that attentional regulation is the critical factor in this negative relationship. In the present study, twelve mothers were videotaped interacting with their thirteen month old children, and 100 maternal utterances were coded for pragmatic intent. Prescriptives were coded as either leading or following the child's previously established focus of attention. The frequency of mothers' follow-prescriptives was positively correlated with a productive vocabulary measure taken at 22 months. These results indicate that directing a thirteen month old's behavior can have beneficial effects on subsequent vocabulary development, if it occurs during periods of joint focus.

8. Mary E. Farmer and Susan E. Bryson

Spatial and Sequential Processing in Dyslexic and Normal Readers

Some researchers claim that the problem dyslexics have with remembering the order of items is specific to phonetically codable material. On the other hand, Tallal (1984) has proposed that the difficulty is with processing any rapidly occurring sequential stimuli. These alternative hypotheses were investigated by presenting three tasks to three groups: dyslexics, age-matched, and reading-matched controls. The three tasks involved reproducing the identity and/or location of four letters which appeared at various rates either simultaneously or sequentially in a matrix, or sequentially in the same spot. Results indicated that dyslexics had more difficulty than age-matched controls with sequential presentation, and their performance deteriorated even more when both identity and location were required. Our findings provide support for the idea that dyslexics suffer a deficit in processing both sequential and phonetically codable material.

9. Florence Rosenthal

The frequency and behavioral manifestation of post-traumatic stress disorder in sexually abused children: A proposal.

The purpose of the proposed study is to assess the frequency and behavioral manifestation of PTSD in school-age children who have been sexually abused, using a representative sample of verified cases. There will be a matched comparison group of physically abused children. The types and prevalence of behavior problems exhibited by sexually abused children will be assessed using the Child Behavior Checklist (Achenback & Edelbrock, 1983). Any problems found will be examined for relationships with characteristics of the abusive event, the child's pre-abuse development, the number and type of negative life events experienced by the child, family characteristics and family support. It is expected that children with high levels of behavioral problems will have had a more severe abuse experience and/or a more difficult pre-abuse history and will be more likely to have symptoms of PTSD.

10. John Barresi

Descartes' mind, Descartes' body: The ontogenesis of a distinction between mind and matter.

It is generally acknowledged that a major turning point in western man's self-understanding occurred when Descartes originated the distinction between mind and matter. Descartes' distinction and his mind/body dualism struck such a responsive chord that we have yet to extricate ourselves from it, as indicated by the existence of cognitive science and neuroscience as separate but interactive disciplines. In the present talk I will try to provide a theoretical account of the causal origin of this distinction through a psychosocial investigation of Descartes' life history and its socio-cultural context. I will also try to account for its long-term consequences.

11. W.K. Honig

Pigeons Discriminate Numbers of People in Pictures

Pigeons readily learn to distinguish slides of natural scenes that contain a person from slides that do not. We asked whether they would also be sensitive to the number of people in a picture. Four birds were trained to discriminate slides containing one person from slides containing "many", where "many" was defined as eight or more. For two birds, one person was positive; for two others, "many" was positive. The birds were then tested with other numerical categories: the presence of 2, 3, 4-5, and 6-7 persons in the picture. All birds provided orderly discrimination gradients among these test values. We asked whether the numerosity discrimination had to be made explicit in training in order to obtain such a gradient. For this purpose, we trained a second group of birds that had to discriminate "many" from the absence of any person in the picture. This was an easy discrimination for them, and a test with the same numerical categories again provided an orderly discrimination gradient. These data provide evidence for number discriminations within "natural" categories, but that is not the same as counting.

12. B.L. Alsop and W.K. Honig

Flashes of red, flashes of blue. If you're a pigeon, what do you do?

Pigeons can discriminate the relative numerosness of one type of stimulus element versus another type when these two types of stimuli are presented simultaneously in a two dimensional array (Honig, last year). Recently we have been investigating pigeons ability to discriminate the relative numerosness of two types of stimuli presented sequentially. Pigeons are presented with 5 "flashes" of light. Some flashes are a blue light, others are a red light. If the sequence contained more blue than red (an extreme example would be 5 blue and 0 red) then pecking the left key may provide a reinforcer, but pecking the right key darkens the chamber for 3 seconds. If there were more red, then these contingencies are reversed. Accuracy in this task appears to be a function not only of the relative number of elements, but also their arrangement within the sequence.

13. Doug Williams and Vin LoLordo

Black is beautiful, but white is alright.

When given a choice, rats prefer spending time in a black compartment of a two-compartment apparatus when the other compartment is painted white and brightly lit. This unconditioned preference for black was used as a baseline against which the effects of various Pavlovian treatments on contextual learning was assessed. Rats were locked in the black compartment and given mild electric shocks. To our surprise they now preferred the white compartment in a follow-up preference test.

14. Anastasia Droungas, Andrew Delamater, and Vincent LoLordo

Is disgust necessary in taste-mediated potentiation of cue aversion

Animals learn aversions to exteroceptive cues, e.g. visual, auditory, olfactory, and contextual cues only if during the poisoning episode these cues are compounded with taste. This phenomenon has been termed taste-mediated potentiation of cue aversion and seems to defy the general principle of stimulus equipotentiality. Garcia has proposed that potentiation is a phenomenon unique to the feeding system since it is observed only in preparations where LiCl is used as the US and not in preparations where shock is used. LiCl is a potent toxicosis inducing agent that also produces shifts in the hedonic evaluation of tastes that have been paired with it. Shock on the other hand, is a fear inducing agent and it does not produce palatability shifts. This talk will explore whether it is necessary to use toxicosis inducing agents in order to produce potentiation.

15. A.R. Delamater & V.M. LoLordo

Expectation, surprise, and event processing

The question of the relationship between how well an event is anticipated and how well it is processed has been answered by students of animal learning and students of human visual orienting in different ways. Whereas investigations of Pavlovian conditioning have supported the claim that surprising events are processed more than expected events, the reverse seems true for investigations of spatial orienting, for example. I'll present data from a couple of experiments which incorporates Pavlovian methodology into what may be considered a "form" orienting paradigm with human subjects. A stimulus (SC+) which signalled the occurrence of the target form (i.e., a letter) seemed to facilitate processing of that target, and another stimulus (CS-) which signalled the absence of the target form seemed to inhibit processing of the target when it was presented on probe trials. I'll discuss the implications of these results for future investigations of Pavlovian conditioning.

16. R.P. Croll and K.G. Bina

Is Phillip Rushton wrong? Can one both have a large brain and be sexually well endowed?

Recent work indicates that the nervous systems of gastropods grow post-embryonically by increasing both the size and the number of resident neurons. For example, the PeIb cluster within the pedal ganglia of the pond snail, *Lymnaea*, is undetectable in juvenile snails. It first appears near the time of sexual maturity and thereafter the cells within the cluster increase rapidly in both size and number. Parallel growth occurs in the cluster's target tissue, the penis. Our future studies will employ both ablation and transplantation experiments to test the role of the penis in determining the growth of the PeIb cluster. We hypothesize that contrary to certain well-established (but perhaps not scientifically supported) beliefs, a large penis and perhaps a well used penis might enhance neural development. If generalized to other animals this hypothesis could have profound social and ethical ramifications.

17. Benjamin Rusak & Harold A. Robertson (Pharmacology)

Foscinating rhythm, I'm under your spell.

We have been studying the neural mechanisms by which lighting cycles entrain (synchronized) circadian (daily) rhythms in mammals. The suprachiasmatic nuclei (SCN) in the hypothalamus serve as the dominant pacemaker of the circadian system, and cells here are responsive to retinal illumination. We have recently found that these cells alter their gene expression in response to photic cues. Brief light exposures cause SCN cells (and not other retinorecipient cells) to increase expression of the cellular-fos (c-fos) gene, a member of the family of proto-oncogenes. This

discovery presents new opportunities for analysing the transmitters in the entrainment pathway, the intracellular events involved in entrainment, and the biochemical mechanism of the circadian clock.

18. J.H. Brandstatter, I.A. Meinertzhagen, and S.R. Shaw

Degenerate Synapses

We are interested in the plastic changes which synapses undergo during normal adult life. In the fly's visual system, for example, synapses formed by the terminals of photoreceptors in the first optic neuropile, or lamina, gradually decrease in number with age. The dynamic events of their loss are being investigated by procuring the death of photoreceptors by photoablation, following treatment with a dye sulphorhodamine. The terminals commence degeneration within one hour, progressing through characteristic changes of mitochondrial swelling, terminal shrinking and cytoplasmic darkening. Their synapses disappear, but asynchronously, progressing through the following sequence: first the presynaptic ribbon disappears, then the postsynaptic elements separate from the presynaptic site (usually together), and finally glia insinuate themselves into the enlarged synaptic cleft.

19. I.A. Meinertzhagen and H. Seyan

Inter-species ocular transplantation in flies: intended synaptic bestiality.

Transplantations of eyes and brains have been undertaken between flies, of the same or different species, with the aim of procuring inter-species synaptogenesis. Our objective is to examine how conserved the intercellular signals that are widely hypothesised to underly synapse formation have been between flies of progressively remote ancestry. Current progress and its implications will be reported.

20. S.R. Shaw

Evolution of the visual optics and associated neural wiring in Diptera.

In pursuit of the nature of evolutionary modification in nervous systems, so far we have looked mostly at neuronal connectivity, finding that there has been wholesale change in the synaptic connections between an invariant set of neuronal homologues, in Diptera (flies). The visual optics have received less attention, but evolved in parallel from a more primitive symmetrical state to form the asymmetric clusters of photoreceptors found exclusively in today's advanced families. This switch apparently happened in a single transition in an ancestor back in the Triassic or earlier. A highly stereotyped projection pattern of axons is needed behind the eye to take advantage of the new optics, raising the question of whether this "neural wiring" changed in step with the optical changes, or occurred earlier or later, out of synchrony. Reconstructions of the axonal projection pattern in several primitive groups indicates a very tight linkage between the optics and the wiring, pointing probably to some kind of coupling between the two seemingly independent developmental programmes. In attempting to shed more light on the transition state, I have started to examine the optics in a primitive group that lies near the stem line of the advanced Diptera, as a prelude to tracing its axonal projection, and will show how the optics can be analysed in vivo with a suitably arranged microscope, and what the results predict so far.

21. P. Kind, D.E. Mitchell and C. Beaver

Hybridoma Technology: A Technique for Identifying the Molecular Basis Developmental Plasticity.

Many hypotheses for the specificity of connections in the nervous system involve the presence of cell surface molecules which act as recognition sites between appropriate neurons. Indeed, many

theories concerning the molecular mechanisms which underly plasticity in developing mammalian pathways include the presence of surface molecules capable of mediating synaptic reorganization. Hybridoma technology provides a potential vehicle for identifying the cellular antigens which underly synaptogenesis and developmental plasticity. In the present study, the visual cortex of kittens reverse sutured at four weeks for four days were used to immunize adult mice in an attempt to produce antibodies specific for developmental antigens of the visual system. We will describe our attempt with the hybridoma procedure and in addition present some results obtained with CAT 301, a monoclonal antibody produced by Dr. Susan Hockfield at Yale University which labels a subset of neurons in the cat visual system.

22. Richard E. Brown

put up poster put in slides

Rearing rats in a germ-free environment eliminates their odours of individuality.

To test the hypothesis that commensal bacteria influence the urinary odours of individuality, we collected urine from PVG and PVG.R1 male rats born by Caesarian section and reared in a germ-free environment. Urine from the germ-free rats was not discriminated in a habituation-dishabituation test, while urine from males kept in a conventional animal house could be discriminated. When the germ-free rats were moved to a conventional animal house their urine was discriminable. These results provide the first evidence that the MHC and commensal bacteria interact in the production of the unique individuality signal in the urine of congenic rat strains. Similar interactions between animals and their commensal bacteria are responsible for producing other social odours.

23. Heather A. Schellinck, Valerie C. Rouse & Richard E. Brown

The use of operant conditioning methods to study discrimination of complex social odours in rats.

Using an habituation-dishabituation task, Brown, Singh & Roser (1987) showed that the urine of rats which differed only in the A region (Class 1) of the Major Histocompatibility Complex (MHC) was discriminable by other rats. To further this research, we constructed a small animal olfactometer so that operant conditioning techniques could be used to test the ability of rats to discriminate between odours. Preliminary results suggest that the procedure is appropriate for demonstrating the ability of rats to discriminate between the odours of two individual male rats of an outbred strain; the odours of congenic strains of rats and human odours. However, inconsistencies in the results would suggest a slack of stimulus control and modifications are now being made before further hypotheses are tested. The nature of these hypotheses will be discussed.

24. Cynthia A. Staicer

Vocal communication systems of Dendroica warblers.

The songs of birds play an important role in intraspecific communication, though mechanisms involved remain poorly understood. Song use in Dendroica warblers suggests different songs serve different functions: Males use A songs when advertising for mates and B songs when interacting with other males. A and B songs differ structurally and are associated with different singing behaviours (slow, repetitive, stereotyped vs. rapid, alternating, variable.) This unusual vocal system presents an opportunity to study underlying mechanisms, including the neural organization of singing behaviours, the control of switching between singing behaviours, and the process of learning songs and their appropriate uses.

25. Elizabeth M. Coscia, Dennis P. Phillips, and John C. Fentress

Moan, groans, screams, and whines, was what she heard from behind the blinds.

The development of the wolf (*Canis lupus*) vocal repertoire was examined in a litter of wolf pups during the first 35 days of life. Audio and video recordings were collected from inside a den located within a 3.8 hectare wooded enclosure. The vocalizations recorded were analyzed spectrographically to trace changes in acoustic content with postnatal age. The neonatal sounds varied in harmonic structure, fundamental frequency, duration, frequency modulation, temporal patterns, and time of first appearance. Harmonically-rich vocalizations present on day one were seen to lower in fundamental frequency with increasing age. Other, often non-harmonic vocalizations, were less common during the first two weeks but by the end of the first month had acoustic structures consistent with their being precursors to adult vocalizations.

26. Catherine L. Ryan and Richard E. Brown

The modification of neonatal ultrasound vocalizations in the rat by norepinephrine agonists and antagonists.

Infant murid rodents emit ultrasonic vocalizations (UV) to guide maternal behavior. Although it has been reported that lesions to the noradrenergic (NE) system do not alter the frequency of UV emitted by rodent pups, we investigated the relationship between NE activity and UV using drugs which act as NE agonists or antagonists. Rat pups were tested at 4, 8, and 12 days of age. Five groups of pups were given a single s.c. injection of one of two doses of clonidine (alpha 2 agonist) or propranolol (beta blocker) or saline. Twenty minutes following injection, activity levels and isolation-induced ultrasounds were assessed using an ultrasonic microphone and a computer compatible ultrasound digitizer. Significant alterations in UV were detected. These results suggest a role for NE in the manifestation of UV and indicate that measurement of neonatal UV can be a sensitive tool for detecting perinatal perturbations of the NE system.

27. Sandra L. Wright and Richard E. Brown

Importance of paternal care in mice when required to forage for food.

Studies of parental behaviour in rodents generally provide the animals with ad lib food and water in a warm environment. No studies have examined parental behaviour when environmental conditions mimic the real world, i.e. food must be foraged and temperature changes accommodated. In this experiment we examined the male's contribution to the pup's development in mice (*Mus musculus*) when wheel running was necessary to obtain food. Two groups of mice were compared: solitary dams with litters (sire absent) and paired dams with litters (sire present). Paired dams weaned more pups than solitary dams but there were no differences in pup weight from birth to weaning. Both groups showed parental care in the light phase and wheel running in the dark phase of the light:dark cycle. However, paired dams alternated parental care and running with the sire. When the sire ran, the dam was on the nest and when the dam ran the sire was on the nest. These results suggest that the benefits of paternal care will be more evident in extreme environmental conditions than in the standard laboratory environment.

28. Richard E. Brown and Lynda A. Wisker

Diet and Genetic differences influence maternal odour preferences of infant mice.

Olfactory signals communicate information about the species, sex, reproductive condition, individuality and maternal state of an animal (Brown, 1979, *Adv. Study Behav.*, 10, 103-162). The

subjects were 16-19 day old mice of two congenic strains (C57BL/6J and B6-ARK-H-2^k), which differ only in the major histocompatibility complex on chromosome 17. We tested the preferences of these mice for maternal odours of lactating females of the two strains on two different diets. Pups were attracted more to odours from lactating females of the same strain and on the same diet as their mother to lactating females of the other strain or those on the other diet. These results indicate that pups may "imprint" on different components of the maternal odour. Dietary information so learned may be used to direct food choices and genetic information to influence the choice of social partners and mates.

29. Maria Legerstee

Communicative Imitation in mother-infant and stranger-infant dyads during the first year.

This study focused on the development of imitation between infants, their mothers and female strangers, and how these behaviors of the participants changed over time. Eight infants were observed longitudinally during the first year of life in natural interactions with the adults. Five behaviors were measured of the interactants: Mouth Open, Eye Open, Lip Protrusion, Smiling, Negative Face and Vocalizing. Lag sequential analyses were performed for each dependent variable. The z-score calculations were subsequently adjusted for the effects of autocorrelations. Significant imitative patterns for all behaviors occurred for the infants and adults at various points throughout development, although the vocalizations of the infants suppressed those of the adults. However, imitative development was not uniform. Peaks in imitative development occurred between 9 and 23 weeks for both infants and adults and between 34 and 45 weeks for infant and mother only. Log linear analyses revealed that significantly more imitative responding took place between mother and infant, than between stranger and infant. Differential imitative responsiveness occurred as early as three weeks for mouth open and vocalizations, and significantly more smiles of the mother were imitated throughout development. The study reveals that imitation is an integral part of infant communication and illustrates the usefulness of this paradigm for investigating the development of imitation during the first year.

30. Jenny Ryon, Richard Brown, and John Fentress

Bringing home the bacon: food provisioning within a captive pack of coyotes (*Canis latrans*)

Social behaviour in a pack of coyotes was observed for six years. This report concerns pup and related feeding by adults. All adult members of coyote packs feed young. Some members also feed pregnant and lactating females and yearlings. In our study, feeding began just before the birth of pups and continued for about 24 weeks. One adult male (Obie) was the only individual to feed the mother as well as the pups in the two years pups were born. The mother, another adult male and adult female also fed the pups in these two years. Obie and the mother but not the other adults also fed the yearlings. All males were vasectomized before the mating season of the third year, thus no further pups were born. Obie continued to feed his mate (the mother in past years) at the appropriate time of year (i.e., just before the expected time of birth, as measured from the time of mating). He also fed his "pups" from previous years. He continued this pattern of feeding for four years (until his death) despite the fact that one of the male "pups" he fed had become dominant to him and one of the female "pups" had become dominate to his mate. The mother also continued to feed older offspring in the absence of new pups. We present data for six years and discuss these with respect to the stimuli which activate food provisioning within coyote packs.

31. April D'Aloisio and Ray Klein

Aging and Attention

This study examined age differences in attention and the relationship among different tasks previously used to measure attention. Sixteen community-dwelling older adults (X age = 61.1) and 16 young undergraduates (X age = 19.4) were tested on three attention tasks: Treisman's serial search task, Eriksen's distractor task and LaBerge's attentional range tasks. For all tasks, stimuli were arranged on an imaginary circle in order to eliminate any effects of retinal eccentricity. In Treisman's task the old showed slower search rates than young which suggests that the old were slower to shift their attention than the young. Both the amount of interference and the shape of the interference function in the Eriksen task was similar for both young and old indicating that there was no age difference in the size of the attentional 'beam'; the probe data from the LaBerge narrow task also support this conclusion. In sum, the results show that older subjects are able to control the size of the attentional 'beam' as well as young subjects, but they may not be as efficient at shifting their 'beam' as the young.

32. Annabel J. Cohen

Context effects on discrimination of complex tones.

As part of a research program examining the effects of small integer frequency ratio relations on memory for unfamiliar (microtonal) sets of tones, discrimination of nine complex tones (tones separated and discriminanda differing by one-third semitone) was tested under simultaneous and successive context conditions. The pattern of benefits of context differed from that obtained in absolute judgement studies of the same stimuli. This sensitivity of context effects to memorial demands of the task assists in solving the puzzle of whether benefits of musical relations are sensory or cognitive in origin.

33. Ray Klein, Glen Berry, Kevin Briand and Barbara D'Entremont

What the eye tells the brain declines with increasing retinal eccentricity; but does it do so differently for good and poor readers?

Geiger & Lettvin answer this question affirmatively in a recent article in the prestigious New England Journal of Medicine. In their study normal readers were more accurate at reporting letters near fixation than dyslexic readers, but at greater eccentricities the dyslexics were better. This pattern survived several methodological improvements imposed by psychophysically minded psychologists (W. Dember et al), suggesting a basic difference in the eye-brain interface. Close scrutiny of both studies reveals several methodological flaws, which leave wide open the possibility that the dyslexic and normal readers use different attentional strategies. Following closely the task used by Geiger & Lettvin and Dember, but improving upon their displays to minimize the possibility of strategy differences, we seek to provide a more definitive answer to our title. Other research bearing on the nature of dyslexia may be described.

34. Alan Kingstone

"Great Expectations" and other tales of attention and set.

Delayed crossing the Atlantic Ocean

Sensitivity or criterion: do ERPs enhance our understanding of the issue?

This interdisciplinary study was concerned with a central issue in visual attention research: does advance knowledge of the spatial location of an impending stimulus facilitate its detection by enhancing early perceptual processing (Posner et al., 1978; 1980) or does expectancy affect decision criteria for responding (Shaw, 1984)? Both event-related potentials (ERPs) and RT performance data were collected in three conditions: simple RT, choice RT, and a second discrimination task that involved the manipulation of stimulus-response likelihood. The RT data show costs and benefits of cuing for the simple, choice and the likely condition of the third task, while the least likely stimulus condition of the third task does not show a cuing effect when validly cued, a replication of Klein and Hansen's (1987) finding of spotlight failure. The onset of ERP enhancements were compared both across the three conditions and between the likely and unlikely conditions of the third task and will be discussed in relation to the issue.

Recognition and recall: Late ERP components in patients with Alzheimer's Disease, MID, alcohol-related disorders and normal controls.

This is a presentation of proposed research that will employ a late positive event-related potential (ERP) component ("P300") as an index of memory processing in aged patients with memory dysfunction and controls. Past research has indicated a positive relationship between amplitude of the "P300" elicited by a word upon initial presentation and the probability of that word being later remembered. In phase 1, ERPs will be recorded in response to words presented during an incidental learning task. Subjects will then be asked to recall as many of these words as possible. In phase 2, using a slightly different list, subjects will be asked to indicate (recognize) whether the word had been presented during Phase 1. The relationship between "P300" amplitude and memory performance will be examined to determine neurophysiological differences between the clinical groups.

"The Return of Dr. Rushton", with the help of Shirley MacLaine

By Tim Jukes

